Sequences and Series: Geometric Sequences

Questions 1–72

Question 1. Write the nth term, u_n, for each of these geometric sequences in the form

$$u_n = ar^{n-1}$$

where a is the first term and r is the common ratio.

a) $3, 12, 48, 192, 768, \ldots$

b) $80, 120, 180, 270, 405, \ldots$

c) $240, 120, 60, 30, 15, \ldots$

d) $10, 9, 8.1, 7.29, 6.561, \ldots$

e) $22, -44, 88, -176, 352, \ldots$

f) $\frac{1}{3}, -\frac{4}{3}, \frac{16}{3}, -\frac{64}{3}, \frac{256}{3}, \ldots$

g) $\frac{9}{10}, -1, \frac{10}{9}, -\frac{100}{81}, \frac{1000}{729}, \ldots$

h) $88, -66, 49.5, -37.125, 27.84375, \ldots$

Question 2. It is given that $1.5, 3, 6, 12, \ldots$ is a geometric sequence. Find the 15th term in the sequence.

Question 3. It is given that $6, 8, \frac{32}{3}, \frac{128}{9}, \ldots$ is a geometric sequence. Find the 6th term in the sequence.

Question 4. It is given that $0.004, 0.012, 0.036, 0.108, \ldots, 57395.628$ is a finite geometric sequence. Determine how many terms there are in the sequence.

Question 5. It is given that $1.02, 4.08, 16.32, 65.28, \ldots, 1069547.52$ is a finite geometric sequence. Determine how many terms there are in the sequence.

Question 6. It is given that $10000, 11000, 12100, 13310, \ldots, 21435.8881$ is a finite geometric sequence. Determine how many terms there are in the sequence.

Question 7. A geometric progression has first term 0.2 and common ratio 2. Find the first term in the sequence that exceeds 1000.

Question 8. A geometric progression has 4th term 28 and 15th term 57344. Find the first term and the common ratio.

Question 9. A geometric progression has 3rd term 2 and 7th term 13122. Find the possible first terms and their corresponding common ratios.

Question 10. A geometric progression has 2nd term 15 and 5th term -234375. Find the first term and the common ratio.

Question 11. A geometric progression has 6th term 62.208 and 7th term 373.248. Find the first term and the common ratio.

Question 12. A geometric progression has 3rd term 19.8 and 6th term 534.6. Find the first term in the sequence that exceeds ten million.

Question 13. A geometric sequence has 2nd term $1 + 2\sqrt{5}$ and 3rd term $38 + 19\sqrt{5}$.

a) Show that the common ratio is $8 + 3\sqrt{5}$

b) Find the 1st term in the sequence, writing it in the form $p + q\sqrt{5}$, where p and q are rational numbers.

Question 14. A geometric sequence has the three consecutive terms $(k + 1)$, $5k$, $\frac{50k}{7}$, where k is a constant. Find the value of k.

Question 15. A geometric sequence has the three consecutive terms $\frac{49k}{36}$, $(k+2)$, $(3k-2)$, where k is a constant. Find the possible values of k.

Question 16. A geometric sequence has the three consecutive terms 2^{3k-7}, 2^{2k+1}, 2^{4k-5}, where k is a constant. Find the value of k.

Question 17. A geometric sequence has the three consecutive terms $3e^{7/4+2k}$, $\frac{15}{2}e^{k+67/24}$, $\frac{75}{4}e^{101/24-k}$, where k is a constant. Find the value of k.

Question 18. Find the sum of the geometric series

$$5 + 10 + 20 + \cdots + 2560$$

Question 19. Find the sum of the geometric series

$$0.13 + 0.52 + 2.08 + \cdots + 2129.92$$

Question 20. Find the sum of the geometric series

$$\frac{1}{5000} + \frac{1}{1250} + \frac{2}{625} + \cdots + \frac{8192}{625}$$

Question 21. Find the sum of the geometric series

$$9000 - 9900 + 10890 - \cdots - 17538.4539$$

Question 22. Find the sum of the geometric series

$$\frac{2}{3} - 1 + \frac{3}{2} - \cdots + \frac{19683}{512}$$

Question 23. Aaliyah decides to go on a seven-day walking holiday along a Highland trail, camping at the end of each day. On the first day, she walks 15 miles. On each subsequent day, she walks 8% less than the previous day.

a) Determine how far she walks on the last day, to two decimal places.

b) Determine the total distance Aaliyah walks on her holiday, to two decimal places.

Question 24. A geometric series is given by

$$\sum_{i=6}^{25} \frac{4}{3} \times (\frac{2}{5})^{i-1}$$

a) Write down the first term and common ratio of the series.

b) Write down how many terms there are in the series.

Question 25. Find the sum of the first 8 terms of the following geometric series to 2 decimal places:

$$66 + 88 + \frac{352}{3} + \cdots$$

Question 26. Find the sum of the first 9 terms of the following geometric series to 2 decimal places:

$$5200 + 4160 + 3328 + \cdots$$

Question 27. A geometric sequence has kth term given by $u_k = 128 \times (\frac{7}{2})^{k-1}$. Show that

$$\sum_{k=1}^{5} u_k = 26840$$

Question 28. A geometric sequence has rth term given by $u_r = 19683 \times (\frac{1}{3})^{r-1}$. Show that

$$\sum_{r=4}^{10} u_r = 1093$$

Question 29. A geometric sequence, u_n, has a common ratio $r > 1$. Find the value of r such that

$$\sum_{n=1}^{4} u_n = 8 \sum_{n=1}^{2} u_n$$

Question 30. A geometric series has first term 100 and common ratio $\frac{9}{10}$. It is given that the sum of the first k terms is greater than 870.

a) Show that

$$k > \frac{\ln 0.13}{\ln 0.9}$$

b) Hence find the smallest possible value of k

Question 31. A geometric series has first term 2 and common ratio $\frac{26}{25}$. It is given that the sum of the first k terms is greater than 1000.

a) Show that

$$k > \frac{\ln 21}{\ln 1.04}$$

b) Hence find the smallest possible value of k

Question 32. A geometric series has first term 20 and common ratio $\frac{1}{8}$. It is given that the sum of the first k terms is less than 22.85.

a) Show that

$$k > \frac{\ln 3200}{\ln 8}$$

b) Hence find the smallest possible value of k

Question 33. Find the smallest possible value of k such that the sum to k terms of the series

$$5 + 10 + 20 + 40 + \cdots$$

exceeds 1500.

Question 34. Find the smallest possible value of k such that the sum to k terms of the series

$$1.1 + 3.3 + 9.9 + 29.7 + \cdots$$

exceeds 30000.

Question 35. A small clothes shop made a net profit of £250 in their first week of trading. The shop's owner predicts that the net profit each week will form a geometric sequence, increasing by 3% each week.

a) Find the net profit predicted by the owner for the end of the fifth week.

b) How many weeks will need to have passed since opening for the weekly net profit to be over £1000?

c) Determine the total amount of net profit in one year (52 weeks) of trading?

Question 36. Explain why the geometric series

$$7 + 14 + 28 + 56 + \cdots$$

is not convergent and therefore has no finite sum to infinity.

Question 37. Determine which of the following geometric series are convergent:

a) $0.4 + 1.2 + 3.6 + 10.8 + \cdots$

b) $10 + 5 + 2.5 + 1.25 + \cdots$

c) $20 + 30 + 45 + 67.5 + \cdots$

d) $100 + 95 + 90.25 + 85.7375 + \cdots$

e) $100 - 95 + 90.25 - 85.7375 + \cdots$

f) $1.6 - 1.44 + 1.296 - 1.1664 + \cdots$

Question 38. Find the sum to infinity of the geometric series with the following properties:

a) $a = 30,\ r = 0.1$

b) $a = 800,\ r = \frac{2}{5}$

c) $a = 99,\ r = 0.99$

d) $a = -8000,\ r = \frac{3}{8}$

e) $a = 410,\ r = -\frac{5}{6}$

f) $a = -9,\ r = -\frac{11}{12}$

Question 39. A geometric series has first term 80 and sum to infinity 90. Find the common ratio r.

Question 40. Find the sum to infinity of the geometric series

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$$

Question 41. Find the sum to infinity of the geometric series

$$100 + 80 + 64 + 51.2 + \cdots$$

Question 42. Find the sum to infinity of the geometric series

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{6}} + \frac{1}{3\sqrt{2}} + \frac{1}{3\sqrt{6}} + \cdots$$

Question 43. A geometric series has first term 50 and the sum to infinity is less than 100. Find the range of possible values of r.

Question 44. A geometric series has first term 9 and the sum to infinity is less than 40. Find the range of possible values of r.

Question 45. A geometric series has first term 100 and the sum to infinity is greater than 500. Find the range of possible values of r.

Question 46. A geometric series has first term 80 and the sum to infinity is greater than 100 but less than 200. Find the range of possible values of r.

Question 47. Find the value of

$$\sum_{n=1}^{\infty} 3000 \times (\frac{2}{3})^{n-1}$$

Question 48. Find the value of

$$\sum_{n=3}^{\infty} 25350 \times (\frac{11}{13})^{n}$$

Question 49. Find the value of

$$\sum_{n=10}^{\infty} 1000000 \times (\frac{1}{5})^{n-1}$$

Question 50. The geometric sequence u_n has terms $18, \frac{27}{2}, \frac{81}{8}, \frac{243}{32}, \ldots$

a) Find the sum to infinity of the series generated by the terms of u_n

b) The sequence v_n is found by multiplying each term in the sequence u_n by 32. Write down the first four terms of the sequence v_n

c) Using just your answer to a), write down the sum to infinity of the series generated by the terms of v_n

Question 51. The 2nd term of a geometric series is 10. The sum to infinity is 40. Find the first term and common ratio.

Question 52. The 2nd term of a geometric series is 10800 and the sum to infinity is 45000. Find the two possible first terms of this series.

Question 53. The 2nd term of a geometric series is 95 and the sum to infinity is $\frac{9500}{9}$. It is given that $r > \frac{1}{2}$. Find the sum of the first 6 terms in the series to two decimal places.

Question 54. A geometric sequence, u_n, has first term a and common ratio r. It is given that

$$\sum_{n=1}^{\infty} u_n = 10 \sum_{n=1}^{2} u_n$$

Find the two possible values of r.

Question 55. The first three terms of a geometric series are given by $\frac{5}{k}$, $\frac{1}{k^4}$ and $\frac{k}{80}$, where k is a positive constant. Show that the sum to infinity can be written in the form $ak+b$, where a and b are rational numbers and k is a value to be found.

Question 56. The first four terms of a geometric series are:

$$10+\frac{10}{2k}+\frac{10}{(2k)^2}+\frac{10}{(2k)^3}+\cdots$$

where k is a positive constant. It is given that the series has a sum to infinity. State the possible values of k.

Question 57. Find the value of

$$\sum_{n=1}^{\infty}(\frac{1}{2}\ln e^2 - 2\ln e^{1/4})^n$$

Question 58. Find the exact value of

$$\sum_{n=1}^{\infty}(\sin 45^\circ \cos 30^\circ)^n$$

Question 59. Find the exact value of

$$\sum_{n=1}^{\infty}(\frac{\sin(90+180n)^\circ}{5})^n$$

Question 60. Find the exact value of

$$\sum_{n=1}^{\infty}(\frac{\sqrt{5}}{2}-\frac{\sqrt{3}}{3})^n$$

Question 61. A geometric sequence has nth term $u_n = \frac{100}{9} \times (\frac{3}{5})^{n-1}$. Solve the inequality

$$\sum_{n=1}^{\infty} u_n - \sum_{n=1}^{k} u_n > 0.09$$

where k is a constant.

Question 62. A geometric sequence has first term a and common ratio r. Show that S_n, the sum of the first n terms, is $\frac{a(r^n-1)}{r-1}$.

Question 63. A geometric sequence has nth term $u_n = 8000 \times (\frac{1}{2})^{n-1}$. Solve the inequality

$$\sum_{n=1}^{\infty} u_n - \sum_{n=1}^{k} u_n < 5000$$

where k is a constant.

Question 64. A geometric sequence has nth term $u_n = k \times 0.5^{n-1}$, where k is a positive constant.

a) Find the value of the 5th term in terms of k

b) Show that the sum to infinity is $2k$

c) Explain how the sum to infinity would be affected if the nth term of the sequence was $v_n = 0.5 + k \times 0.5^{n-1}$

Question 65. Jenny is an art student. She decides to paint a series of thin concentric circles on a canvas, with the first being the largest, having a radius of 40 cm, and the next one always has a radius that is 98% the length of the previous one. The total length of all the circumferences of the circles on the canvas will not exceed k cm. Find the value of k.

Question 66. A geometric sequence has first term 108 and common ratio $\frac{3}{8}$. Show that the nth term for the sequence can be written as

$$\frac{3^{n+2}}{2^{3n-5}}$$

Question 67. A geometric sequence has first term 400 and common ratio $\frac{5}{4}$. Show that the nth term for the sequence can be written as

$$\frac{5^{n+1}}{2^{2n-6}}$$

Question 68. A geometric sequence has first term $(2k^2 + 5k - 6)$ and second term $(2k + 5 - \frac{6}{k})$, where k is an irrational constant. It is given that the sum to infinity is 3. Find the value of k.

Question 69. It is given that

$$\sum_{n=1}^{\infty}(\frac{k+1}{2k+3})^n = 2$$

Find the value of

$$\sum_{n=1}^{6}(\frac{k+1}{2k+3})^n$$

Question 70. Prerequisite knowledge for Q70–Q72: Recurrence Relations & Arithmetic Sequences
Determine if any of these sequences are arithmetic, geometric, or neither:

a) $u_{n+1} = u_n + 3,\ u_1 = 5$

b) $u_{n+1} = 2u_n,\ u_1 = 4$

c) $u_{n+1} = 2u_n + 3,\ u_1 = 6$

Question 71. It is given that u_n is an arithmetic sequence with first term 3 and common difference 4. It is also given that v_n is a geometric sequence with first term 2 and common ratio 3.

a) Determine whether the following sequence is arithmetic, geometric or neither:

$$(u_1 + v_1), (u_2 + v_2), (u_3 + v_3), \ldots, (u_n + v_n)$$

b) Find the exact value of

$$\sum_{n=1}^{8} (u_n + v_n)$$

Question 72. An arithmetic progression has first term 10 and common difference 700, and a geometric progression has first term 5 and common ratio 3. The kth term for both sequences is the first time when the geometric term is greater than the arithmetic term.

a) Show that $3^k + 414 > 420k$

b) Determine the value of k